

April 4, 2024, 12:30 to 1:45 pm. Open book, open notes. No electronics devices.

(10) Question 1. You have a system that measures parameter x . You are given the range as 0 to x_{max} (2cm), and a signal to noise ratio as SNR in dB (60 dB). Explain how to use these parameters to choose the fixed-point system to represent x . Give the answer and the explanation.

Type the integer (signed or unsigned):

Unsigned because x is unsigned

Resolution:

$$SNR = 20 \log_{10}(x_{max}/noise)$$

$$Precision \text{ in alternatives} = x_{max}/noise = 10^{SNR/20} = 1000$$

$$EOB = \log_2(1000) = 10 \text{ bits}$$

$$Resolution = x_{max}/precision = 2cm/1000 = 0.002cm \text{ (anything smaller is ok)}$$

Fixed point resolution typically is 10^n or 2^m , so 10^{-3} or $2^{-9}cm$ are possibilities

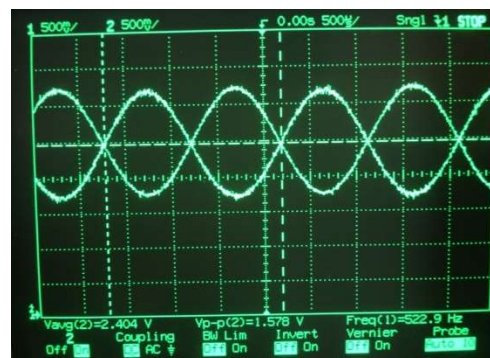
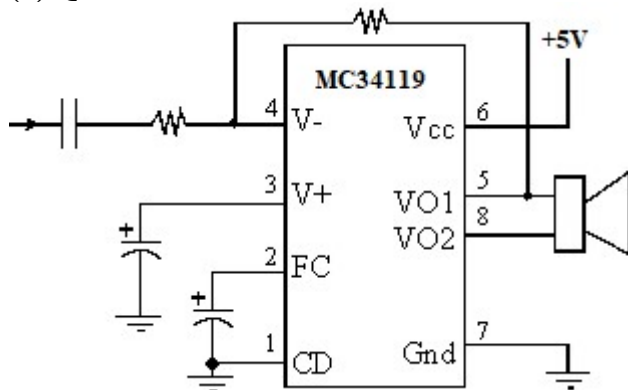
Precision the integer (8 16 or 32 bits):

Smallest integer will be 0

Largest integer will be 1000 $\rightarrow 1000 \cdot 0.002cm = 2cm$

16-bit unsigned integer

(5) Question 2. Circle all the reasons to drive a speaker with the MC34119 versus with a rail-to-rail op amp.



More power to the speaker

Higher voltage difference across the speaker

Op amps have higher voltage gain

Op amps have higher frequency response

Both operates 0 to 5V

Both block DC component, only passing AC sound

(25) Question 3. There is a master microcontroller using SPI sending data about once every 1ms. One 8-bit data transmission requires about 2 μ s to complete. The SPI clock is approximately 2.5 MHz. The PB2 PB1 PB0 labels show the pin assignments on your slave TM4C123. These three pins are not supported by hardware SPI in the TM4C123, so you will have to implement the interface using simple GPIO input/output.

(5) Part a) What event should trigger an interrupt on your slave TM4C123? For example: “interrupt on both the rise and fall of PB1”, or “interrupt periodically at 1ms”. These are examples, neither is correct.

400ns/12.5ns is 32 bus cycles, time for 16 instructions. Fall of PB0 because 400 ns is too short to process 8 separate interrupts. If the clock were much slower you could interrupt on falling edge of PB2 (when data is stable) Interrupting on the rising edge of PB2 is always wrong because you would be reading data when it is not stable

(15) Part b) Write the C code for the ISR in your slave that receives the data and puts each data byte into a software FIFO. Reading PB1 returns 2 or 0.

```
void GPIOPortB_Handler(void) {
    GPIO_PORTB_ICR_R = 0x07;          // acknowledge interrupt
    uint32_t n = 0;
    uint32_t data = 0;
    for(n=0; n<8; n++){
        while(PB2){}; // wait for fall
        data = (data<<1) + (PB1>>1); // read input
        while(PB2==0){}; // wait for rising
    }
    FIFO_Put(data);

    uint32_t n = 0;
    uint32_t data = 0;
    while(PB0 == 0) { // more like how it actually works
        while(PB2){}; // wait for fall
        data = (data<<1) + (PB1>>1); // read input
        while(PB2==0){}; // wait for rising
    }
    FIFO_Put(data);

    uint32_t data = 0;
    while(PB2){}; // wait for fall
    data |= (PB1<<6); // read bit 7
    while(PB2==0){}; // wait for rising
    while(PB2){}; // wait for fall
    data |= (PB1<<5); // read bit 6
    while(PB2==0){}; // wait for rising
    ...
    while(PB2){}; // wait for fall
    data |= (PB1>>1); // read bit 0
    while(PB2==0){}; // wait for rising
    FIFO_Put(data);
}
```

Part c) The real SPI interrupts on rising edge of PB0, end of frame

(10) Question 4. There are three 32-bit binary fixed-point numbers. All three are I27.Q5, meaning the value of the number is the integer times 2^{-5} . You may assume that **i**, **j**, and **k** are declared in C with `int32_t i,j,k`; This software can only access **i,j,k** with no other variables allowed. Write C code that multiplies the first number times the second number and stores the product in the third number.

```
// value of first = i/32
// value of second = j/32
// value of third = k/32
// third = first*second
// k/32 = i/32 * j/32
k = (i*j)>>5;

// FYI addition
// k/32 = i/32 + j/32
k = i+j;
// FYI division
// k/32 = i/32 / (j/32)
k = (i<<5)/j;
```

(15) Question 5. Design an analog circuit with the following transfer function: $V_{out} = V_1 + 2*V_2$. You may assume inputs V_1 and V_2 are such that output range will be 0 to 3.3V. You may use any chips shown in the book or presented in class. Show your work and label all chip numbers and resistor values. You do not have to show pin numbers. You do not need to add a low pass filter. Use E12 values

10	12	15	18	22	27	33	39	47	56	68	82
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Table 9.1. E12 Standard resistor and capacitor values for 5% tolerance.

No offset needed, so no reference needed

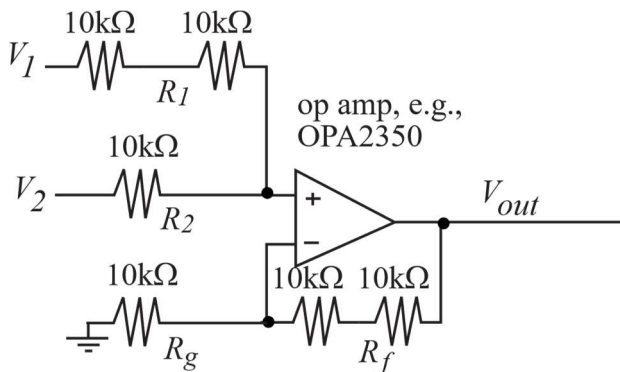
Add ground gain $V_{out} = V_1 + 2*V_2 - 2*V_g$. so sum of gains is 1

Gain1 = 1

Gain2 = 2

GroundGain = 2

Notice there are no E12 resistors with a 2/1 ratio, so use two resistors in series (or two in parallel)



(10) Question 6. Let `uint32_t x[8]` be eight ADC data values sampled at $f_s=8$ kHz. Derive an equation to calculate the magnitude of frequency component of the input at $f=2$ kHz

Δf is $f_s/8 = 1$ kHz

Index at 2kHz is $k = 2$

$W = \exp(-j2\pi/8)$

W^n are the 8 points on the unit circle

$W^0 = 1$

$W^2 = -j$

$W^4 = -1$

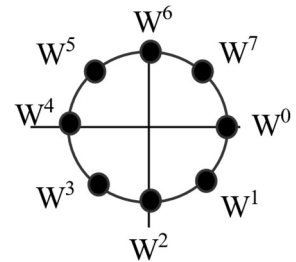
$W^6 = j$

$A2 = \text{sum}(x[n] * W^{2n})$ from $n=0$ to $n=7$

$\text{ReA2} = x[0] - x[2] + x[4] - x[6];$

$\text{ImA2} = -x[1] + x[3] - x[5] + x[7];$

$\text{MagA2} = \text{sqrt}(\text{ReA2} * \text{ReA2} + \text{ImA2} * \text{ImA2});$



(15) Question 7. You are given a 1000 mA-hr 12V battery. A 100% efficient buck regulator converts the 12V battery input to the 3.0V supply for the microcontroller system. The bus clock of the microcontroller is reduced to 1 MHz to save power. To save even more power, the software runs in two modes. Sleep mode draws 10mA, and active mode draws 50 mA. The active mode runs for X percent of the time (%). The remaining 100-X percent is in sleep mode. What is the maximum percent X (in percent) you could run if the system needs to run for 100 hours? Show your work.

Power budget 1000mA-hr > Y*100hr,

Y is average battery current = 10mA

Since the regulator is 100% efficient, battery power = microcontroller power

10mA battery current will supply up to $12/3 * 10\text{mA} = 40\text{mA}$ at the microcontroller

$X * 50 + (100 - X) * 10 = 4000$

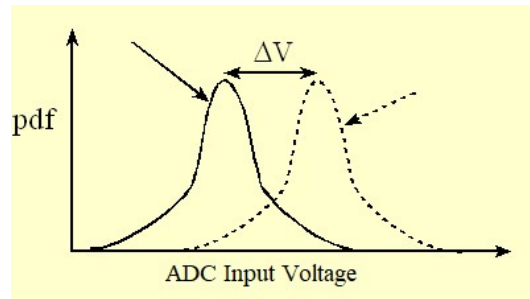
$40X = 3000$

$X = 75\%$

(10) Do Q8 or Q9, not both

Question 8. Explain how to use the **student's t-test** to experimentally determine ADC resolution. Draw figures to explain.

Collect N measurements at a constant input V . Change input to $V + \Delta V$ and take another N measurements. Use student's t test to determine if the two sample sets are equal. If we reject the hypothesis, then the ADC resolution is less than or equal to ΔV .



Question 9. Explain how the **Central Limit Theorem** applies to a measurement system.

Part a) What must you assume to apply the CLT?

Assume noise is random with same distribution

Assume noise is samples is independent of other samples

Not officially necessary, but nice if noise has 0 mean

Part b) What must software do to improve the SNR of the measurement?

Take multiple samples and sum/average